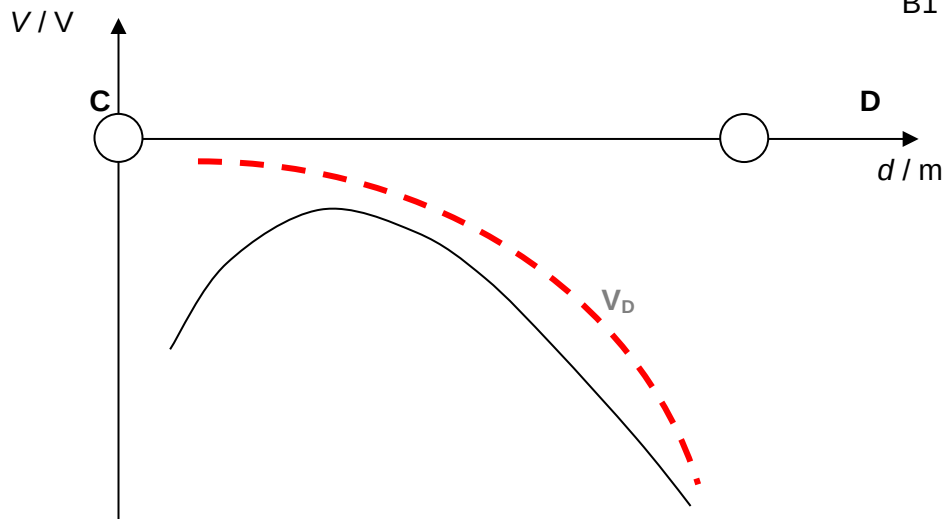


6 (a)



B1

- (b) **C** and **D** are both negatively charged as the total electric potential between the line joining their centres are negative for all points between them. B1

- (c) The total potential at a point between them is the scalar sum of the individual potentials. B1

Maximum potential occurs at **P**, where the scalar sum of the potentials is a maximum. B1

- (d) Electric field strength at point **P** due to **C** alone,

$$E_C = \frac{Q_C}{4\pi\epsilon_0 x^2}$$

Electric field strength at point **P** due to **D** alone,

$$E_D = \frac{Q_D}{4\pi\epsilon_0 y^2} \quad \text{B1}$$

At point **P**, resultant electric field = 0 B1

$$\Rightarrow E_C = E_D$$

$$\Rightarrow \frac{Q_C}{x^2} = \frac{Q_D}{y^2}$$

$$\Rightarrow \frac{Q_C}{Q_D} = \frac{x^2}{y^2} \quad \text{B1}$$